EX. NO: 3

IMPLEMENTATION OF HASH FUNCTION – SHA-1

AIM:

To implement the SHA-1 hashing technique using a Java program.

DESCRIPTION:

SHA-1 (Secure Hash Algorithm 1) is a cryptographic hash function that takes a message of length less than 2^64 bits and produces a 160-bit (20-byte) message digest. It is designed so that it is computationally infeasible to find two different input messages that hash to the same digest. A hash function like SHA-1 is used to compute a fixed-size alphanumeric string (a digest / fingerprint) that represents a file or piece of data, and it is meant to be unique and non-reversible. Note that SHA-1 is now considered cryptographically weak (practical collisions have been demonstrated) and is retained here only for lab/educational purposes — SHA-256/SHA-512 should be used in real applications.

ALGORITHM:

STEP-1: Read the input message from the user.

STEP-2: Obtain a MessageDigest instance for the algorithm "SHA1".

STEP-3: Pad the message internally as per the SHA-1 specification (handled automatically by the library).

STEP-4: Divide the padded message into 512-bit blocks; each block updates a 160-bit internal state (five 32-bit words A, B, C, D, E) over 80 rounds using SHA-1's logical functions and rotations.

STEP-5: Call digest() to obtain the final 160-bit hash value and print it in hexadecimal.

PROGRAM: (Java)

```
import java.security.*;
import java.util.Scanner;

public class SHA1
{
    public static void main(String[] a) throws Exception
    {
        MessageDigest md = MessageDigest.getInstance("SHA1");
        System.out.println("Message digest object info: ");
        System.out.println(" Algorithm = " + md.getAlgorithm());
        System.out.println(" Provider  = " + md.getProvider());

        Scanner sc = new Scanner(System.in);
        System.out.print("\nEnter the input string: ");
        String input = sc.nextLine();

        md.update(input.getBytes());
        byte[] output = md.digest();

        System.out.println("SHA1(\"" + input + "\") = " + bytesToHex(output));
    }

    public static String bytesToHex(byte[] b)
    {
        char hexDigit[] = {'0','1','2','3','4','5','6','7','8','9','A','B','C','D','E','F'};
        StringBuffer buf = new StringBuffer();
        for (int j=0; j<b.length; j++) {
            buf.append(hexDigit[(b[j] >> 4) & 0x0f]);
            buf.append(hexDigit[b[j] & 0x0f]);
        }
        return buf.toString();
    }
}
```

SAMPLE OUTPUT:

```
Message digest object info:
 Algorithm = SHA1
 Provider  = SUN version 21

Enter the input string: run
SHA1("run") = DF6AD19037C97987C4FF9792810C0E145356717C

=== Code Execution Successful ===
```

RESULT:

Thus the SHA-1 hashing technique was implemented successfully using Java and the digest was verified against the standard test vector.